

ECON345 Homework 5: Plots, Dates, and Strings

Due: October 1st, 11:59 PM

As with last week, submit both your R Markdown file (`week_05.Rmd`) and its knitted PDF. Comment your code to explain what you're doing — partial credit rewards a reasonable, well-documented attempt, not just a correct final answer.

Part I: Du Bois and Data Visualization (20 pts)

At the 1900 Paris Exposition, W. E. B. Du Bois and his team presented over 60 charts and statistical tables on the economic and demographic condition of African Americans — some of the earliest and most inventive American data visualization on record. We saw a few of these plates in Lecture 1; this week we'll analyze one and build our own.

Question 1 (4 pts)

Review the pictogram below.

VALUE OF LAND OWNED BY GEORGIA NEGROES.



Source: Library of Congress. *Value of land owned by Georgia Negroes*. <https://www.loc.gov/pictures/collection/anedub/item/2013650437/>

- **[2 pts]** Identify the variables presented in the figure.

- **[2 pts]** What does the figure communicate about the data? Is there a relationship between the variables? Explain.

Question 2 (2 pts)

Consider the chart types covered in Lecture 5. What is another type of data visualization that would effectively convey the information in the figure above, and how?

Question 3 (4 pts)

Based on your answer to Question 2, describe what your new chart would look like.

- **[2 pts]** Explain how you would use the variables to construct the new plot.
- **[2 pts]** Explain what it would communicate to the viewer.

Question 4 (10 pts)

The data behind the chart is `land.csv`.

- **[2 pts]** Import the data into R.
- **[8 pts]** Create a chart from the imported data. If possible, produce the chart you described in Question 3.

```
land <- read_csv("hw/data/land.csv")
```

Part II: Visualizing Wages and Inflation (45 pts)

Question 5 (15 pts)

Create a bar chart visualizing average wage by occupation.

- Load the CPS1985 dataset (`data("CPS1985")`).
- Use `group_by()` and `summarize()` to compute the summary statistic you need.
- Pipe (`%>%`) the result directly into `ggplot`, or save it to a new data frame first.

Question 6 (5 pts)

Which occupation has the highest average wage? Which has the lowest? What are the advantages and disadvantages of comparing groups using the mean, as opposed to another statistic?

Question 7 (10 pts)

Load the `economics_long` dataset from the `ggplot2` library (`data("economics_long")`). Using the methods from lecture, describe at least two key characteristics of this dataset – refer to the lecture for examples of what counts as a “characteristic.”

Question 8 (15 pts)

Create a line plot visualizing inflation, as measured by U.S. Personal Consumption Expenditures (PCE), from 2007 onward. (Hint: `filter()` on date and on `variable == "pce"`.) Use the plot to answer:

- In what year did PCE drop, and why do you think that happened?
- What is the most recent value shown?

Part III: Dates (20 pts)

`economics_dates.csv` stores the same set of months, encoded eight different ways (`date1`–`date8`).

Question 9 (10 pts)

Read in `economics_dates.csv`. Using `mutate()` and the appropriate `lubridate` function for *each* column, overwrite `date1`–`date8` by converting them into Date objects (instead of character or numeric). Consult the lecture slides if you get stuck on which `lubridate` function to use for a given format. Confirm your conversions with `glimpse()`.

```
economics_dates <- read_csv("hw/data/economics_dates.csv")
```

Question 10 (10 pts)

Now that `date1`–`date4` are proper Date objects, use `mutate()` and `format()` to change how each one is *displayed*, matching the style below. Confirm your results with `select()` and `head()`.

- `date1`: Feb-1967
- `date2`: February 1, 1967
- `date3`: 1967.02.01
- `date4`: 2/1/67

BONUS: Strings (20 pts)

Run the following chunk so the story text loads into your environment. `base` or `stringr` functions both work for these exercises (`stringr` will probably be easier).

```
story <- read_lines("hw/story.txt")
```

- **[10 pts]** Find every **word** starting with a vowel (capitalized or not) and count how many times each one occurs. You'll need the regular-expressions material from lecture to complete this.
- **[10 pts]** Count how many times each punctuation symbol occurs in the story.

Survey

How long did it take you to complete this assignment? Your response will not affect your grade.

Submission

Upload two files to Canvas before the deadline:

- your `week_05.Rmd`, and
- the **knitted PDF**.

As always: comment your code, and if you get stuck, show your work. If your `.Rmd` won't knit, get each piece working before you move on to the next, and restart R and knit clean if results look stale.